

6. Optimization: Choosing the Next Test

Construct new candidates, measure their weakness, and use the results to guide further search.

Start from twinning-selected medoid sets

INITIAL CANDIDATES

Twinning selects K observed customer identities per set.

Declared starts create the initial population.

MEASURED RECORDS

Assign customers to each fixed set.

Record regional contrast, coverage and support.

FURTHER SEARCH

Choose a parent set.

Propose another valid set.

Evaluate it and update the records.

Measured feedback determines which candidates are constructed after the initial assessment.

Change candidate identities within fixed rules

FIXED DURING THIS SEARCH

Predictor, error trees and leaf distance.

Allowed customer pool and K.

Discovery sample and conditions.

Loss, support and reporting rules.

CHOSEN BY THE OPTIMIZER

Which medoid set to explore.

Which identity to replace.

Which allowed identity to insert.

When to try a fresh set.

Each proposed set defines a new partition; its medoids stay fixed while it is evaluated.

Replace one medoid, then assign and measure

PARENT c_0

Medoids {A, D}.

Choose A for replacement.

Choose allowed identity B.

CHILD c_1

Medoids {B, D}.

Check distinct identities and the required set size.

EVALUATION

Assign every discovery customer to B or D.

Measure both resulting regions.

Store the new scores with c_1 . Each region keeps its medoid identity and parent candidate ID.

Sharper contrast can come with lower coverage

Same 1,000 customers; show one region from each partition. All regions are scored.

Record	Region / rest Brier	Contrast	Count	Coverage
c_0 / A	0.0900 / 0.0200	0.07	100	10%
c_1 / B	0.1098 / 0.0198	0.09	80	8%

Both satisfy the example rule: at least 50 observations per side and coverage $\geq 8\%$.

Retain both tradeoffs. The declared preference for highest eligible contrast selects c_1 / B.

Choose a search method for the candidate space

Method	How the next test is chosen
Designed sweep	Follow a predeclared batch of candidates.
Hill climbing	Test valid neighbors; move to the best improvement.
Evolutionary search	Select parents; construct and evaluate offspring.
Bandit allocation	Allocate another trial to an existing option.
Bayesian optimization	Fit a score surrogate; choose by its acquisition rule.

Use the same evaluation contract and cost accounting when comparing search methods.

Retain evidence, allocate search, select a finding

RETAIN

Keep useful contrast-coverage tradeoffs and their records.

Example: keep c_0 / A and c_1 / B.

ALLOCATE

Choose which parent set gets another attempt.

Use a declared selection rule and exploration policy.

REPORT

Apply the final reporting rule.

Example: choose the sharpest eligible region, then freeze it.

Customer coverage is a regional property. Selection probability controls how search spends its budget.

Give each condition a declared share of attention

Four fixed region records, one per parent set. Entries are Brier contrast against each sample's rest.

Record	Normal	High rates	Recession	Pop. shift
q ₁	0.042	0.030	0.055	0.020
q ₂	0.025	0.061	0.058	0.071
q ₃	0.018	0.022	0.031	0.074
q ₄	0.035	0.044	0.040	0.033

Use $\delta = 0.005$ as a declared near-tie rule; assess sampling uncertainty separately.

Split condition credit among shortlisted records

Each condition allocates 0.25, split among records within δ of its maximum.

Condition	Maximum – δ	Shortlist	Credit per record
Normal	0.042 – 0.005	q_1	0.250
High rates	0.061 – 0.005	q_2	0.250
Recession	0.058 – 0.005	q_1, q_2	0.125 each
Pop. shift	0.074 – 0.005	q_2, q_3	0.125 each

Total ownership shares: $q_1 = 0.375$; $q_2 = 0.500$; $q_3 = 0.125$; $q_4 = 0$.

All four records are Pareto-nondominated across conditions. Ownership share is an allocation rule.

Mix focused allocation with exploration

Selection probability = $0.95 \times \text{ownership share} + 0.05 / 4$.

Record	Ownership share	Selection probability
q_1	0.375	36.875%
q_2	0.500	48.750%
q_3	0.125	13.125%
q_4	0	1.250%

Draw $u = 0.90$ falls in q_3 's cumulative interval (0.85625, 0.98750).

Select q_3 's parent medoid set, construct a valid move and measure the child's scores.

Change a factor level, then rerun the tests

PARENT: P₂ FROM DECK 3

Unverified \$50 anchor.

Direct wording.

Irrelevant background present.

Task: retrieve the approved fee.

CHILD: CHANGE THE WORDING

Keep the \$50 anchor and background.

Use the declared paraphrased wording.

Render the same fee task.

Evaluate all declared conditions.

Preserve the task and reference policy. Score new responses and record any unscorable cases.

Count the work and declare when search ends

ACCOUNT FOR EVALUATIONS

Predictive: assignments and regional scores for every proposed set.

Agentic: responses, tool calls and scorer runs across conditions.

Record invalid proposals and retries.

DECLARE THE STOPPING POLICY

Set a maximum evaluation budget.

Define any early-stop improvement rule.

Retain the search history and best eligible records.

Apply the final reporting rule.

For $K = 4$, 100 newly evaluated partitions produce 400 single-cell score attempts per condition.

Freeze the selected rule and confirm it

DISCOVER

Construct → assign → score.

Retain useful tradeoffs.

Use feedback for further candidates.

FREEZE

Select the eligible finding.

Save medoids and region identity.

Freeze measurement and condition rules.

CONFIRM

Apply the saved rule to reserved evidence.

Report contrast, coverage, support and uncertainty.

The search uses discovery outcomes. Confirmation evaluates the final frozen selection.

A new construction rule opens a different search

CURRENT REGIONAL RULE

Choose a medoid set.
Assign customers to its cells.
Evaluate each cell as a region.
Search further medoid sets.

POSSIBLE STRUCTURAL EXTENSION

Permit a region to combine two cells.
Check and admit that membership rule.
Search under the revised rules.
Measure the resulting tradeoffs.

Deck 7: use discovery evidence to propose, admit and evaluate changes to what candidates can express.